

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Artificial Intelligence and Data Science
ASSESSMENT TOOL 1 – Pattern Recognition and Text Processing

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	Assessment:	Assessment Tool 1 – Pattern Recognition and Text Processing
Weightage:	40% of CIA		
CO5 Statement:	Analyse regular expression patterns. (BL-3)		
Student Name:	A.Syed Thoufiq	Reg. No.:	192424211
Section / Batch:	Slot D / 2024	Date:	19/7/26

Code of Conduct

I A Syed Thoufiq(192424211) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: A Syed Thoufiq

Instructions

- Show all intermediate steps, calculations, state transitions, and reasoning wherever applicable.
- Use only the Python re (Regular Expressions) module to solve the given problems.
- Clearly mention the Regular Expression (Regex) pattern used before writing the corresponding Python program.
- Display the required output for each question, including extracted values, validation results, counts, and positions wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Regular Expression Pattern Generation	Pattern Matching Information Extraction	1
Search for a String	Keyword Search and Text Analysis	2
Split the documentation into sentences and words	Text Tokenization and Sentence Segmentation	3
Email and Strong Password Validation	Data Validation and Format Verification	4
Compile Once, Validate Many	Pattern Reusability and Performance Optimization	5

Name : A Syed Thoufiq

Reg No : 192424211

Course/Code : DSA0308

Date : 10/07/26

QUESTION 1: REGULAR EXPRESSION PATTERN GENERATION

Code:

```
import re
```

```
text = """
```

```
Student Details:
```

```
Name: John Doe
```

```
Email: john.doe123@gmail.com
```

```
Mobile: +91-9876543210
```

```
Password: P@ssw0rd123
```

```
Date of Birth: 15/08/2004
```

```
Register Number: 23AIML1056
```

```
Department: Artificial Intelligence and Machine Learning
```

```
"""
```

```
# Regular Expression Patterns
```

```
email_pattern = r'[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}'
```

```
mobile_pattern = r'\+91-\d{10}'
```

```
password_pattern = r'[A-Za-z0-9@#$$%^&*!]+'
```

```
dob_pattern = r'\d{2}/\d{2}/\d{4}'
```

```
register_pattern = r'\d{2}[A-Z]{4}\d{4}'
```

```
print("Email ID:")
```

```
print(re.findall(email_pattern, text))
```

```
print("\nMobile Number:")
```

```
print(re.findall(mobile_pattern, text))
```

```
print("\nPassword:")
```

```
password = re.search(r'Password:\s*(S+)', text)
```

```
print(password.group(1))
```

```
print("\nDate of Birth:")
```

```
print(re.findall(dob_pattern, text))
```

```
print("\nRegister Number:")
```

```
print(re.findall(register_pattern, text))
```

Output:

```
PS C:\Users\ASHARAF\Desktop\DSA0308- NLP> & C:/Users/ASHARAF/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/ASHARAF/Desktop/DSA0308- NLP/Q1-AT1.PY"
Email ID:
['john.doe123@gmail.com']

Mobile Number:
['+91-9876543210']

Password:
P@ssw0rd123

Date of Birth:
['15/08/2004']

Register Number:
['23AIML1056']
PS C:\Users\ASHARAF\Desktop\DSA0308- NLP>
```

QUESTION 2: SEARCH FOR A STRING

Code:

```
import re
```

```
paragraph = """
```

```
Artificial Intelligence (AI) is transforming industries across the world.
AI is used in healthcare to assist doctors in diagnosis,
banking to detect fraud, and in education to provide personalized learning experiences.
Many companies invest heavily in AI research because AI improves efficiency
and enables intelligent decision-making.
As AI continues to evolve, professionals with AI skills are in high demand.
```

```
"""
```

```
word = "AI"
```

```
match = re.search(word, paragraph)
```

```
if match:
```

```
print("First occurrence found.")
print("Starting Position :", match.start())
print("Ending Position  :", match.end()-1)
```

```
total = len(re.findall(word, paragraph))
print("Total Occurrences :", total)
```

```
else:
```

```
print("Word not found.")
```

Output:

```
PS C:\Users\ASHARAF\Desktop\DSA0308- NLP> & C:/Users/ASHARAF/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/ASHARAF/Desktop/DSA0308- NLP/Q2-AT1.py"
First occurrence found.
Starting Position : 26
Ending Position  : 27
Total Occurrences : 6
PS C:\Users\ASHARAF\Desktop\DSA0308- NLP>
```

SECTION 3: SPLIT THE DOCUMENTATION INTO SENTENCES AND WORDS

Code:

```
import re
```

```
paragraph = """
```

```
Artificial Intelligence (AI) is transforming industries across the world.
```

```
AI is used in healthcare to assist doctors in diagnosis,
```

```
banking to detect fraud, and in education to provide personalized learning experiences.
```

```
Many companies invest heavily in AI research because AI improves efficiency
```

```
and enables intelligent decision-making.
```

```
As AI continues to evolve, professionals with AI skills are in high demand.
```

```
"""
```

```
print("Original Paragraph:\n")
```

```
print(paragraph)
```

```
# Split into sentences
```

```
sentences = re.split(r'[.!?]+' , paragraph)
```

```
print("\nSentences:")
```

```
count = 1
```

```
for sentence in sentences:
```

```
    sentence = sentence.strip()
```

```
    if sentence:
```

```
        print(f"{count}. {sentence}")
```

```
    count += 1
```

```
# Split into words
```

```
words = re.split(r'\s+' , paragraph.strip())
```

```
print("\nWords:")
```

```
print(words)
```

```
print("\nTotal Sentences :", len([s for s in sentences if s.strip()]))
```

```
print("Total Words :", len(words))
```

OUTPUT:

```
PS C:\Users\ASHARAF\Desktop\DSA0308- NLP> & C:/Users/ASHARAF/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/ASHARAF/Desktop/DSA0308- NLP/Q3-AT1.py"
Original Paragraph:

Artificial Intelligence (AI) is transforming industries across the world.
AI is used in healthcare to assist doctors in diagnosis,
banking to detect fraud, and in education to provide personalized learning experiences.
Many companies invest heavily in AI research because AI improves efficiency
and enables intelligent decision-making.
As AI continues to evolve, professionals with AI skills are in high demand.

Sentences:
1. Artificial Intelligence (AI) is transforming industries across the world
2. AI is used in healthcare to assist doctors in diagnosis,
banking to detect fraud, and in education to provide personalized learning experiences
3. Many companies invest heavily in AI research because AI improves efficiency
and enables intelligent decision-making
4. As AI continues to evolve, professionals with AI skills are in high demand

Words:
['Artificial', 'Intelligence', '(AI)', 'is', 'transforming', 'industries', 'across', 'the', 'world.', 'AI', 'is', 'used',
', 'in', 'healthcare', 'to', 'assist', 'doctors', 'in', 'diagnosis,', 'banking', 'to', 'detect', 'fraud,', 'and', 'in',
', 'education', 'to', 'provide', 'personalized', 'learning', 'experiences.', 'Many', 'companies', 'invest', 'heavily', 'i',
n', 'AI', 'research', 'because', 'AI', 'improves', 'efficiency', 'and', 'enables', 'intelligent', 'decision-making.', '
As', 'AI', 'continues', 'to', 'evolve,', 'professionals', 'with', 'AI', 'skills', 'are', 'in', 'high', 'demand.']

Total Sentences : 4
Total Words : 59
PS C:\Users\ASHARAF\Desktop\DSA0308- NLP>
```

SECTION 4: EMAIL AND STRONG PASSWORD VALIDATION

Code:

```
import re

email = input("Enter Email ID: ")
password = input("Enter Password: ")

email_pattern = r'^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.(com|org|edu|in)$'

password_pattern = r'^(?=.*[a-z])(?=.*[A-Z])(?=.*\d)(?=.*[@#$$%^&*!]).{8,}$'

# Email Validation
if re.fullmatch(email_pattern, email):
    print("\nEmail ID is Valid")
else:
    print("\nEmail ID is Invalid")

# Password Validation
if re.fullmatch(password_pattern, password):
    print("Strong Password")
else:
    print("Weak Password")
```

OUTPUT:

```
PS C:\Users\ASHARAF\Desktop\DSA0308- NLP> & C:/Users/ASHARAF/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/ASHARAF/Desktop/DSA0308- NLP/Q4-AT1.py"
Enter Email ID: john.doe123@gmail.com
Enter Password: P@ssw0rd123

Email ID is Valid
Strong Password
PS C:\Users\ASHARAF\Desktop\DSA0308- NLP> █
```

SECTION 5: COMPILE ONCE, VALIDATE MANY

Code:

```
import re

emails = [
    "john@gmail.com",
    "student123@yahoo.in",
    "invalidmail.com",
    "admin@college.edu",
    "abc@xyz",
    "nlp_user@gmail.com"
]

pattern = re.compile(r'^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.(com|org|edu|in)$')

print("Email Validation Result\n")

for email in emails:
    if pattern.fullmatch(email):
        print(email, "-> Valid")
    else:
        print(email, "-> Invalid")
```

OUTPUT:

```
PS C:\Users\ASHARAF\Desktop\DSA0308- NLP> & C:/Users/ASHARAF/AppData/Local/Microsoft/WindowsApps/python3.13.exe "c:/Users/ASHARAF/Desktop/DSA0308- NLP/Q5-AT1.py"
Email Validation Result

john@gmail.com -> Valid
student123@yahoo.in -> Valid
invalidmail.com -> Invalid
admin@college.edu -> Valid
abc@xyz -> Invalid
nlp_user@gmail.com -> Valid
PS C:\Users\ASHARAF\Desktop\DSA0308- NLP> █
```

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Q. No. / Criteria	Conceptual Understanding	Accuracy of Answer	Application of Knowledge	Completeness of Response	Clarity & Presentation	Sub. Total	Total %
1	3	3	3	4	4	17	81
2	3	3	3	3	4	16	
3	3	3	3	3	4	16	
4	3	3	3	3	4	16	
5	3	3	3	3	4	16	

Faculty In-charge	Course Coordinator	Program Director
Dr.T.Kumaragurubaran		
Signature: T.Kumaragurubaran	Signature: _____	Signature: _____
Date: 19/7/26_____	Date: _____	Date: _____